



Topic

1. Setting up large data examples
2. Setting Up Sample Apps That Consume Large Data
3. Observing the Performance of Applications That Consume Large Data
4. Observing the Impact of Data Erasers on Application Performance

Purpose

1. Students are able to prove for themselves the impact of data size on application performance.
2. Students are able to prove for themselves the impact of data deletion on application performance.

Introduction

Data warehouse is a concept of managing large amounts of data that accumulate over time to produce a wide variety of valuable information.

In this first practicum, we will prove together the **significant impact of large amounts of data** on the performance of the application/system.

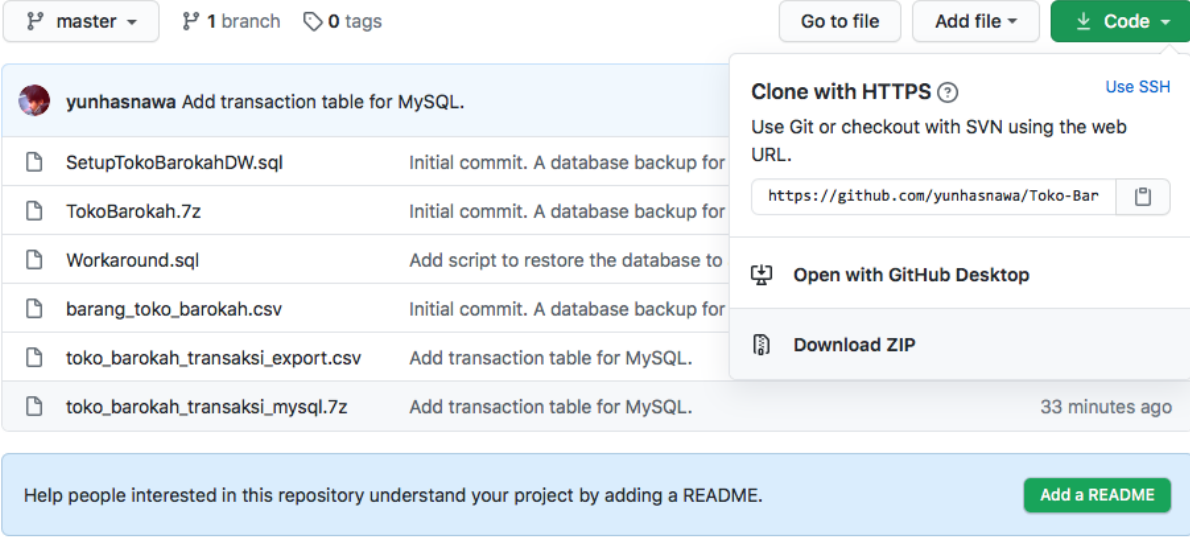
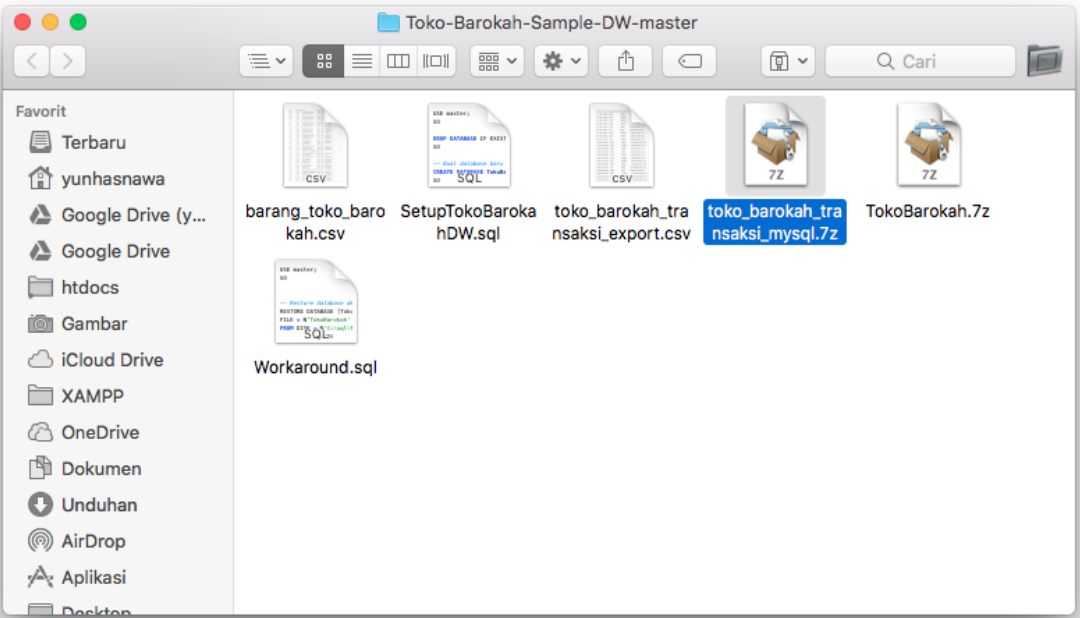
Practicum – Part 1: Setting Up Large Database Examples

In this section, we will first prepare a sample database that will be used for this practicum. Make sure before you start The following steps are installed on your computer:

- PHP
- Web server (Apache, Nginx, etc.)
- MySQL
- phpMyAdmin

You can install the software individually or at once by installing XAMPP.

Step	Information
1	Clone or download SQL files from the following GitHub repository: https://github.com/yunhasnawa/Toko-Barokah-Sample-DW

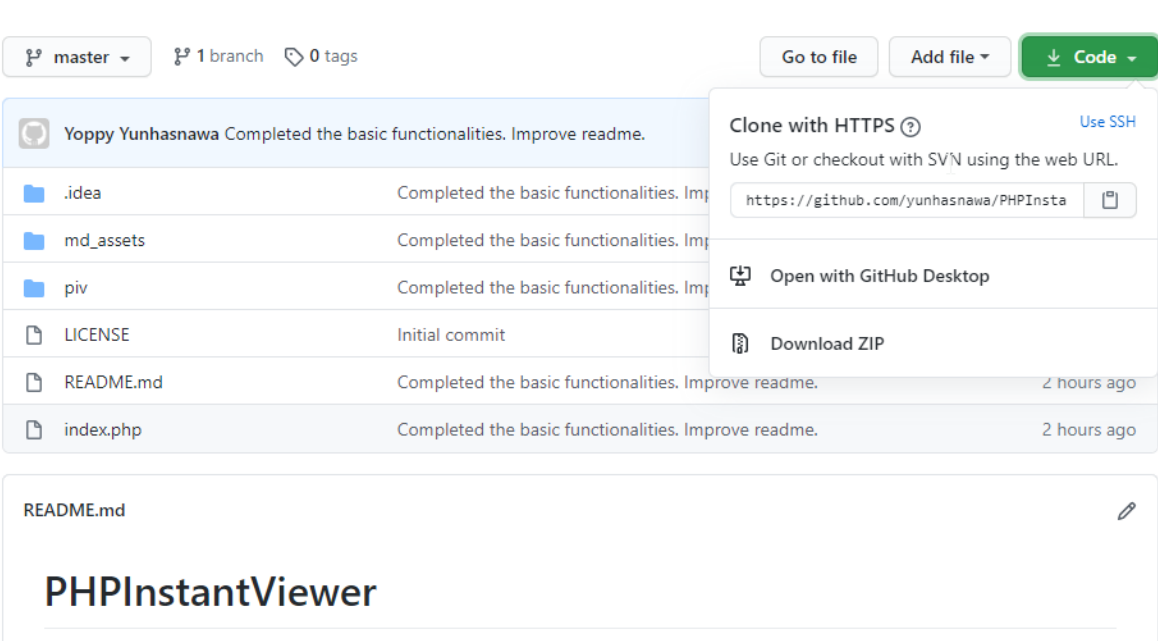
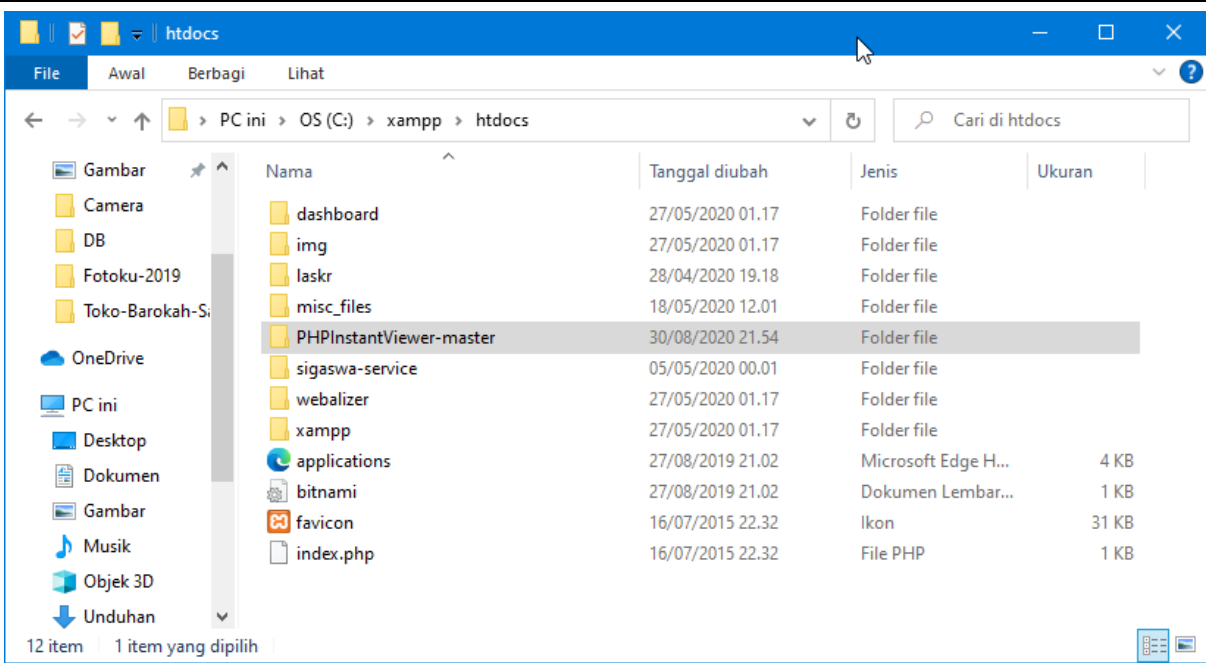
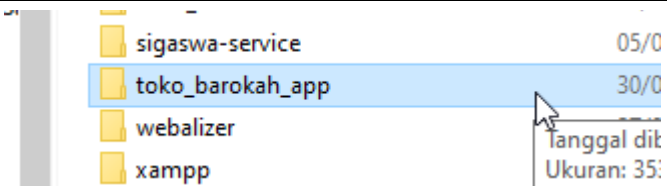
	
2	After the downloaded file is extracted, there will be several files. Extract files with toko_barokah_transaksi_mysql.7z name. You can ignore the other files for a while. 
3	After the 7zip file is extracted, a SQL file named toko_barokah_transaksi_mysql.sql will be generated. Open the file with plain editor text, copy-paste the contents into PHPMysqlAdmin on your computer. Then execute the SQL file. Attention: This SQL file is very large, there will be a chance that it will fail to execute in PHPMysqlAdmin due to insufficient laptop RAM. If this happens, try to execute using CMD as in the following example: https://www.tutorialspoint.com/run-sql-file-in-mysql-database-from-terminal If you still can't, then you can work with your friends whose laptops have high specs.

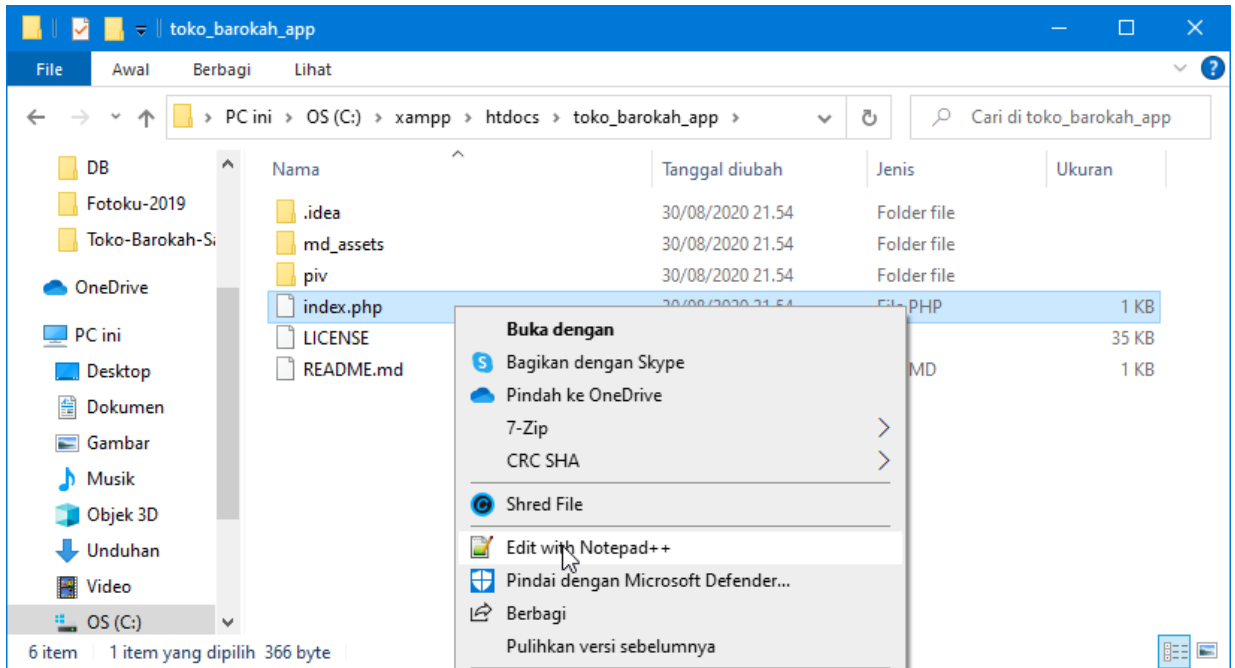
4	If the SQL execution is successful, a new database named toko_barokah will appear on your MySQL server. You can click refresh in the left panel of phpMyAdmin. If it doesn't appear, just refresh your browser. In the database, there is only 1 table with the name of the transaction. In this table there are more than 800,000 lines of transaction data taken from real cashier information systems.

Practicum – Part 2: Preparing an Application Sample

In this practicum section we will try to simulate a web-based application that consumes data from a large database. In this section we use a simple application called **PHPInstantViewer**. We will set the application to display data from the **transaction** table in the **toko_barokah** database that we have prepared beforehand.

Step	Information
1	Clone or download the source code from the PHPInstantViewer app at the following URL: https://github.com/yunhasnawa/PHPInstantViewer

	
2	Extract the downloaded file to your htdocs directory. 
3	Rename the folder that was originally PHPInstantViewer-master to toko_barokah_app. 
4	Open the toko_barokah_app folder. Inside the folder should contain some PHP files. Edit the file named index.php using your favorite text editor.

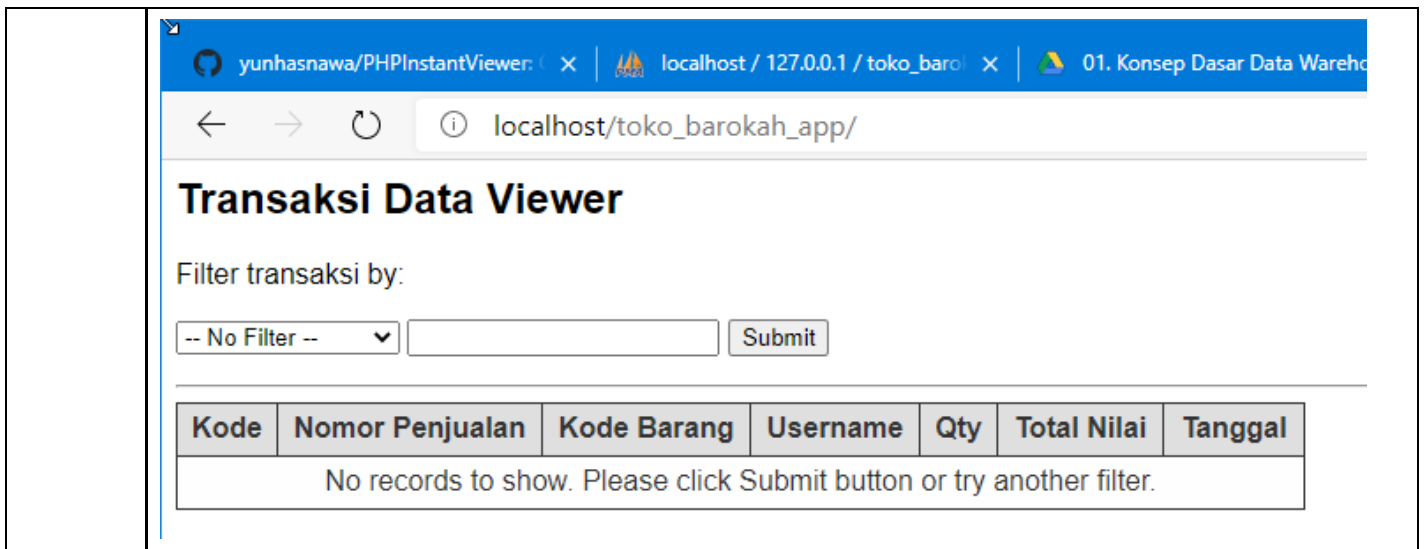


In the file, replace the values of the variables at the beginning of the file as in the blocked section in the following image. Save those changes.

```
*C:\xampp\htdocs\toko_barokah_app\index.php - Notepad++
Berkas Edit Cari Tampilan Encode Bahasa Pengaturan Alat Makro Jalankan Plugin Jendela ?
no_definer.sql new 1 hosts index.php
1 <?php
2
3 require 'piv/Db.php';
4 require 'piv/App.php';
5 require 'piv/View.php';
6
7 use piv\Db;
8 use piv\App;
9
10 // Change these variables to meet your needs
11 $mysqlHost = 'localhost';
12 $user = 'root';
13 $password = '';
14 $dbName = 'toko_barokah';
15 $tableName = 'transaksi';
16
17 $db = new Db($mysqlHost, $user, $password, $dbName);
18 $app = new App($db, $tableName);
19 $app->run();
20
21
```

Open your favorite browser and type in the following URL: http://localhost/toko_barokah_app

If the above steps are done correctly, it will appear as shown in the following image.



Practicum – Part 3: Observing Application Performance with Large Amounts of Data

By this step, everything you need is ready. Next we will see how the application that we have installed performs earlier.

Step

Information

1

This **toko_barokah_app** application is a web-based application that is created using PHP and runs on a local computer. Imagine that the application forgets the same application that was used in the Barokah Store since a few years ago.

Don't press 'Submit' first..

Select **Date** on the combo box filter. Then fill in the following date: **2009-11-17**. Then click the **Submit** button. If the application and database have been installed correctly, then it should not be long before transaction data will appear on that date.

Is it long? No, no..

yunhasnawa/PHPInstantViewer: x

localhost / 127.0.0.1 / toko_barokah_app x

01. Konsep Dasar Data Ware

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localhost/toko_barokah_app/

Transaksi Data Viewer

Filter transaksi by:

Tanggal ▾

2009-11-17

Submit

Kode	Nomor Penjualan	Kode Barang	Username	Qty	Total Nilai	Tanggal
16532	JC-091150715	2064	DEWI	1	42200	2009-11-17
16533	JC-091171041	4116	RIANTI	1	206550	2009-11-17
16534	JC-091150722	1348	DEWI	2	79750	2009-11-17
16535	JC-091150722	7256	DEWI	1	79750	2009-11-17
17056	JC-091171030	2048	DIANTI	1	27150	2009-11-17

Now select **Username** in the combo filter box, then fill in **the Username** and click the **Submit** button.

Unlike the previous one, this time the web page will take more time to display the data. This is indicated by a busy indicator that continues to rotate for more than 30 seconds (depending on your computer's performance).

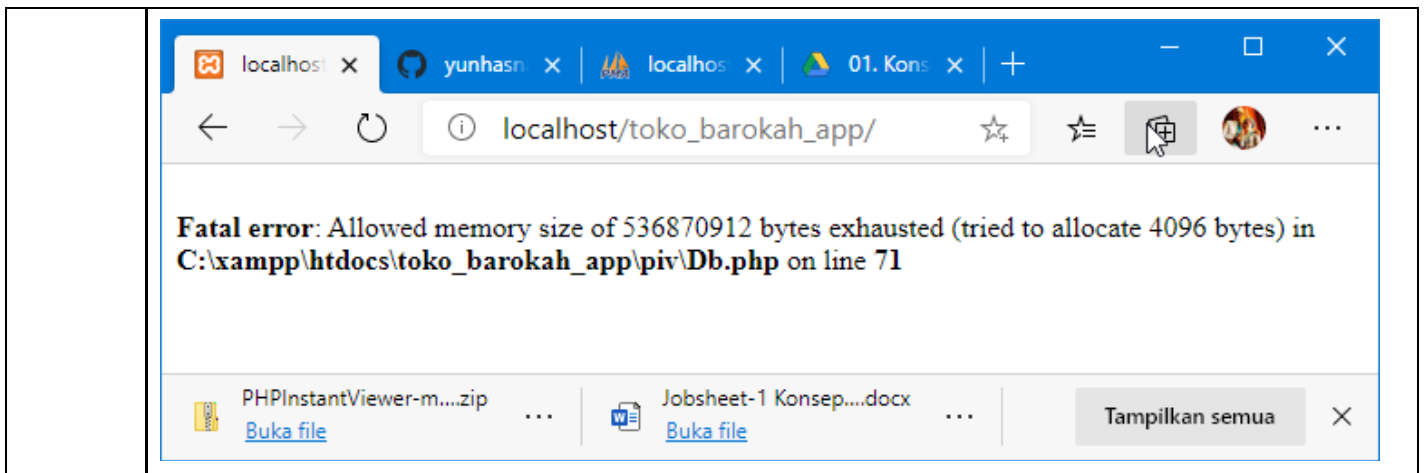
Kode	Nomor Penjualan	Kode Barang	Username	Qty	Total Nilai	Tanggal
2	JC-091170029	6503	RIANTI	1	416925	2009-11-02
3	JC-091170029	5139	RIANTI	2	416925	2009-11-02
4	JC-091170029	2924	RIANTI	1	416925	2009-11-02
7	JC-091071532	4055	RIANTI	2	94300	2009-10-27
8	JC-091071503	1305	RIANTI	1	98500	2009-10-27

Then lastly, do not select anything in the combo box filter (**No. Filter**) and do not fill anything in the textbox to the left of the Submit button.

Click the **Submit** button. What happened?

Kode	Nomor Penjualan	Kode Barang	Username	Qty	Total Nilai	Tanggal
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Since no filter is selected, the application will prompt **all** the rows in the transaction table to be displayed by the MySQL engine. This causes the data to take longer to appear in the browser, and even (depending on your computer's specifications) the application will crash due to running out of memory.



Training Tasks

Create a short report that includes an explanation of the results (along with *screenshots*, if possible) of the following instructions:

1. Note the time it takes for the application to complete the request in the practicum **part-3 step-2**. Starting from the moment you click the Submit button until the *loading indicator* stops.
2. **Remove** from the transaction table, all data that dates back to **2017**. **Note its SQL syntax** in the report.
3. Repeat the 3-step-2 **part-3 practicum** again and also record the time. Compare with the results of **Exercise Task number 1**.
4. Also repeat **the 3-step part-3 practicum**. Compare it to when the data hasn't been deleted.
5. Write down what conclusions you came to after working on this jobsheet on the report.

-- *Good Luck* --